

ADAPTED CAUCHY–LIOUVILLE–MIRIMANOFF POLYNOMIALS AND THE TATE–OORT GROUP SCHEME $\mathbb{TO}_n$ FOR $n \geq 1$

YIMING LIAO AND MILES REID

ABSTRACT. We construct the Tate–Oort group schemes $\mathbb{TO}_n$ for square-free n and for prime power $n = p^r$, extending what we know about the single-prime construction of $\mathbb{TO}_p$ in [TOp]. The main new tool is adapted Cauchy–Liouville–Mirimanoff (*aCLM*) polynomials which are the most simple modification of $(x+y)^n$ to achieve divisibility by n . We will analyze their fibre group schemes, and discuss the blueprint and bottleneck for $\mathbb{TO}_n$ in general under the same method in the final section.

CONTENTS

1. Introduction	1
1.1. History and Background	1
1.2. Narrative Summary	2
2. The CLM and aCLM polynomials	3
2.1. Case for $n = p$ a prime	3
2.2. Case for $n = pq$ with two different primes	4
2.3. The prime power case $n = p^r$	5
2.4. Alchemy polynomials AL_{p^r} for prime powers $n = p^r$	7
2.5. The aCLM polynomial for general n	8
3. Construction of $\mathbb{TO}_n$	10
3.1. Case $n = p$ a single prime	10
3.2. Case $n = pq$ the product of two primes	10
3.3. Case when n is square-free	12
3.4. Case $n = p^r$, particularly p^3	13
3.5. Blueprint for general n	15
References	16

1. INTRODUCTION

1.1. History and Background. In 1839 Cauchy and Liouville [CL] published a positive referees’ report on a single-page paper by Gabriel Lamé [La] proving the $p = 7$ case of Fermat’s last theorem. The referees observe that one step of Lamé’s argument can be seen in a more general framework using the fact that, for a prime $p \geq 5$, the polynomial $(x+y)^p - x^p - y^p$ is divisible by

$$(1.1) \quad \begin{aligned} & pxy(x+y)(x^2+xy+y^2) \quad \text{if } p = 6m+5 \quad \text{or} \\ & pxy(x+y)(x^2+xy+y^2)^2 \quad \text{if } p = 6m+7. \end{aligned}$$

In papers around 1900, the Russian-Swiss mathematician Dmitry Mirimanoff re-worked some of these ideas, again in the context of FLT. See Pavlos Tzermias [Tz] for a more recent discussion. The quotient of $(x+y)^p - x^p - y^p$ by p is the classical *Cauchy–Liouville–Mirimanoff (CLM) polynomial*.

The second-named author [TOp] used the CLM polynomial for a different purpose: to construct a group scheme $\mathbb{T}\mathbb{O}_p$ for a single prime p over an explicit arithmetic base $\mathcal{B}_p = \text{Spec} \mathbb{Z}[t, S]/(St^{p-1} + p)$ that interpolates between the multiplicative group scheme μ_p (where t is invertible) and a Galois twist $\mathcal{C}_p$ of the constant cyclic group $\mathbb{Z}/p$ (where S is invertible), degenerating at the single point $t = S = 0$ to the nonreduced additive group scheme α_p . This gives “the cyclic group $\mathbb{Z}/p$ as you have never seen it before”: a deformation family in mixed characteristic at p joining μ_p to the étale additive cyclic group $\mathcal{C}_p$, via their common degeneration to α_p .

1.2. Narrative Summary. This paper is a continuation of [TOp]. The goal of the series of papers is to extend the results of Frans Oort and John Tate [TO] and [TOp] for prime p to a cyclic group scheme $\mathbb{T}\mathbb{O}_n$ of order n for all $n \geq 1$.

Our main new ingredient is the adapted CLM polynomial introduced in Lemma 2.7. This simple-minded trick adapts $(x+y)^n$ to make all its coefficients divisible by n . More explicitly, we use the Cancellation Lemma (Lemma 2.2) to show that the alternating sum of $(x^{a_I} + y^{a_I})^{b_I}$ is divisible by n , where a_I is a square-free divisor of n :

$$(1.2) \quad F_n = \frac{1}{n} \left(\sum_I (-1)^{\#I} (x^{a_I} + y^{a_I})^{b_I} \right) \in \mathbb{Z}[x, y].$$

Our construction is elementary, and parallel to Möbius inversion for the Euler phi function $\varphi(n) = \prod_{p|n} (1 - \frac{1}{p})$ in basic number theory. It is somewhat surprising that this construction seems not to have been used before. Some sanity checks are also given at the end of each section in Magma code [Ma]

We carry this out in two directions:

- **Square-free n :** we adjoin variables S_I for each nonempty subset I of prime factors of n to the base ring $B = \mathbb{Z}[t, \{S_I\}]/(\{P_I\})$, and the resulting hypersurface defined by the polynomial $F = x^n - \sum_I S_I f_I$ make $\tau = 1 + tx$ the n -th root of unity. The case $n = pq$ with p, q different primes is the model computation, and general square-free case follows the same pattern via an explicit combinatorial identity (Lemma 3.3) for the alternating sums g_J .

- **Prime power $n = p^r$,** with p^3 as the representative case: we introduce variables $S_i = -p^i/t^{p^r-p^{r-i}}$ for $i = 1, \dots, r$ and show that they assemble into a single polynomial $F = x^{p^r} - \sum_i S_i x^{p^i} f_i$ realizing $(1 + tx)^{p^r} \equiv 1$. We identify the fibre group scheme on each stratum of the arithmetic toric base scheme explicitly—over each stratum, it is the product of μ_i and a twist $\mathcal{C}_{p^{r-i}}$ of the cyclic group $\mathbb{Z}/p^{r-i}$ according to which of $t, S_1, \dots, S_r$ is invertible.

In short, with the arithmetic toric base scheme $\mathcal{B}_n$ in place, our construction of $\mathbb{T}\mathbb{O}_n$ is a locally principal hypersurface subgroupscheme $\mathbb{T}\mathbb{O}_n \subset G_{\mathcal{B}_n, t}$ in the hybrid group $G_{\mathcal{B}, t}$ of [TOp, 2.1], with local equations based on the adapted CLM polynomials of Section 2. As in [TOp], we set up local coordinates t_i, S_i on the affine toric pieces of $\mathcal{B}_n$, and the local equations of $\mathbb{T}\mathbb{O}_n \subset G_{\mathcal{B}, t}$ in their affine coordinate rings to achieve the n -torsion condition $(1 + tx)^n = 1$.

However, we should note that this direct approach by defining $\mathbb{T}\mathbb{O}_n$ as a hypersurface does not extend to the construction of $\mathbb{T}\mathbb{O}_n$ for general n . One obstruction is that we have not found a natural way to assemble the prime-power pieces $\mathbb{T}\mathbb{O}_{p^r}$ into a product construction, as we do in the square-free case. More fundamentally, the construction of $\mathbb{T}\mathbb{O}_{pq}$ is already unsatisfactory: the local environments of characteristic p and characteristic q cannot coexist, and so the resulting global object fails to have some properties one would expect from $\mathbb{T}\mathbb{O}_p$. Our attempt to realize a “fibre product-like” group in a single hybrid group $G_{\mathcal{B},t}$ may simply be the wrong framework. Nevertheless, the adapted CLM polynomial is worthwhile in its own right for its elegant form, and we regard this outcome as a useful, if bittersweet, lesson for the sequel.

2. THE CLM AND ACLM POLYNOMIALS

Start from the *ground ring* $\mathbb{Z}$, with the *hybrid variable* t adjoined, that changes gears between multiplicative (when t is invertible), and additive (where $t = 0$). For given n , we view t as the $\varphi(n)$ th root of an unsplit T , that would appear for $n = 1$. In particular, the construction $\mathbb{T}\mathbb{O}_p$ in the prime case $n = p$ of [TOP] has base ring $\mathcal{B}_p = \mathbb{Z}[S, t]/(St^{p-1} + p)$, that is, the subring $\mathbb{Z}[t, p/t^{\varphi(p)}]$ of the Laurent polynomial ring $\mathbb{Z}[t, t^{-1}]$. The case $\mathbb{T}\mathbb{O}_{p^r}$ has S -variables starting with $S_1 = -p/t^{p^r - p^{r-1}}$, where the exponent is $\varphi(p^r)$. In general, we may need to take further roots of t , for example, to pass from n to nm , or to analyse the general case $n = \prod p_i^{r_i}$ in terms of its localisation or completion at the prime factors p_i .

The base scheme $\mathcal{B}_n$ for $\mathbb{T}\mathbb{O}_n$ is an arithmetic toric surface with a birational toric morphism to the affine line $\text{Spec}\mathbb{Z}[t]$. We may view it as a certain iterated blowup at the closed points $V(t, p_i)$ where $n = \prod p_i^{r_i}$.

To the base rings of $\mathcal{B}_n$ we then add a variable x as coordinate function on the group $\mathbb{T}\mathbb{O}_n$. The initial point is to set up a polynomial $f_n(x, t)$, derived from the aCLM polynomial for n , to arrange that $(1 + tx)^n - 1$ is in the ideal generated by $f_n(x, t)$ together with the relations $S_d t^{\varphi(d)} + d$ defining the base rings of $\mathcal{B}_n$. We first illustrate how this works in simple model cases.

2.1. Case for $n = p$ a prime. This case is treated in detail in [TOP]. Over the ground ring $\mathbb{Z}$, the base ring is $B = \mathbb{Z}[t, S]/(St^{p-1} + p)$, with its spectrum $\mathcal{B}_p = \text{Spec}(B)$ as the base scheme. The relation $P = St^{p-1} + p$ sets $S = -p/t^{p-1}$, allowing us to substitute $p \mapsto -St^{p-1}$. One checks that this, together with the degree p monic polynomial in x

$$(2.1) \quad F(x) = x^p - S f_p(x, t) \quad \text{where} \quad f_p(x, t) = \sum_{i=1}^{p-1} \frac{\binom{p}{i}}{p} t^{p-i-1} x^{p-i} \in B[t].$$

provides (as a tautology) the identity

$$(2.2) \quad (1 + tx)^p - 1 = t^p F(x) + \text{multiple of } P.$$

In fact, the product $t^p F(x)$ has first term $(tx)^p$ then (once we swap $t^p S \mapsto -pt$ using P), it gives t times the intermediate terms $i = 1, \dots, p-1$ of $(1 + tx)^p$.

The identity $(1 + tx)^p = 1$ in the coordinate ring is what we need for the hypersurface $F = 0$ to define a p -torsion subgroup of the hybrid group $G_{B,t}$.

2.2. Case for $n = pq$ with two different primes. We use this as an enlightenment to the general square-free cases

Proposition 2.1. *For $n = pq$ with p, q distinct primes, the polynomial*

$$(2.3) \quad (x + y)^n - (x^p + y^p)^q - (x^q + y^q)^p + (x^n + y^n)$$

has all its coefficients divisible by n . Taking out n gives a monic symmetric polynomial $F_n = x^{n-1}y + \cdots + xy^{n-1}$.

This is the simplest modification of $(x + y)^n$ with all coefficients divisible by n . The formula (2.3) is used in 3.2 to construct the Tate–Oort group $\mathbb{T}\mathbb{O}_{pq}$. Here is a baby illustration for $n = 15$: we write out the binomial expansion of $(1 + x)^{15}$, followed by $-(1^3 + x^3)^5$ and $-(1^5 + x^5)^5$, then finally $(1^{15} + x^{15})$:

$$\begin{array}{rcccccc} 1 + 15x + 105x^2 + 455x^3 + 1365x^4 + 3003x^5 + 5005x^6 + 6435x^7 + \text{sym.} & & & & & \\ -1 & & - & 5x^3 & & - & 10x^6 & & - \text{sym.} \\ -1 & & & & - & 3x^5 & & & - \text{sym.} \\ +1 & & & & & & & & + \text{sym.} \end{array}$$

The coefficients $\binom{15}{3} = \binom{15}{12} = 455$ and $\binom{15}{6} = \binom{15}{9} = 5005$ in the first row are not divisible by 3. Subtracting the second row cures this. Also $\binom{15}{5} = 3003 = \binom{15}{10}$ are not divisible by 5, and the third row cures this. However, we subtracted the first and last terms 1^{15} and x^{15} twice, in the second and third rows, and the final row repairs this to take them to 0.

Proposition 2.1 follows from the general result:

Lemma 2.2 (Cancellation Lemma). *Let p be a prime. Then the binomial coefficients modulo p satisfy*

$$(2.4) \quad \binom{pn}{pm} \equiv \binom{n}{m} \pmod{p} \quad \text{for all integers } n, m \geq 1.$$

Moreover, $\binom{p^r n}{pm} \equiv \binom{p^{r-1}n}{m} \pmod{p^r}$.

Proof. Both sides are zero if $m > n$, so assume $m \leq n$. The left-hand side is the fraction

$$(2.5) \quad \frac{\prod_{i=1}^{pm} (p(n-m) + i)}{\prod_{i=1}^{pm} i}.$$

The numerator and denominator are both products of pm terms. Separate them into terms with i coprime to p , of which there are $(p-1)m$, and the m terms with $i = pj$.

As to the former, the numerator is $\prod_{p \nmid i} (p(n-m) + i)$ and the denominator $\prod_{p \nmid i} i$, so term by term, the factors in the two products are congruent mod p . Now for the terms with $i = pj$. In the numerator, they give

$$(2.6) \quad \prod_{j=1}^m p(n-j) = p^m \times \prod_{j=1}^m (n-j).$$

In the denominator, they give $\prod_{j=1}^m pj = p^m \times \prod_{j=1}^m j$. In the fraction the powers of p cancel out, leaving exactly the formula for $\binom{n}{m}$.

The idea for the general prime power p^r is the same. Write out $\binom{p^r n}{pm}$ and get

$$(2.7) \quad \frac{\prod_{i=1}^{pm} (p^r n - i)}{\prod_{i=1}^{pm} i}.$$

Still separate them into terms with i coprime to p and terms with $i = pj$. The terms with $p \mid i$ is the same as above, giving precisely $\binom{p^{r-1}n}{m}$, while to the former the numerator is $\prod_{p \nmid i} (p^r n - i)$ and the denominator is $\prod_{p \nmid i} i$. Modulo p^r and cancel out, and this results in $(-1)^{(p-1)m}$.

So we need only to consider the case when $p = 2$ and m is odd. Here uses a famous lemma by Kummer on binomial coefficient in 1852 as follows: given integers $n \geq m \geq 0$, the p -adic valuation of $\binom{n}{m}$ is equal to the number of carries when m is added to $n - m$ in base p . In this case, the last two digits of $pm = 2m$ is 10 in base 2, so the number of carries when $2m$ is added to $2^r n - 2m$ is at least $r - 1$. As a result, $2^{r-1} \mid \binom{2^r n}{2m}$ and we have

$$\binom{2^r n}{2m} \equiv -\binom{2^r n}{2m} \equiv \binom{2^{r-1}n}{m} \pmod{2^r}$$

as required. $\square$

Remark 2.3. *As an experimental fact, the congruence of Lemma 2.2 holds modulo p^2 for every p and modulo p^3 for $p \geq 5$. For ease of mind, this Magma code [Ma] runs in the online calculator:*

<http://magma.maths.usyd.edu.au/calc/>

```

for p in [2,3] do p;
  &and[IsDivisibleBy(Binomial(p*n,p*m)-Binomial(n,m), p^2)
    : n,m in [2..50] | m le n]; end for;
for p in [5,7,11,13,17,19,23,29] do p;
  &and[IsDivisibleBy(Binomial(p*n,p*m)-Binomial(n,m), p^3)
    : n,m in [2..50] | m le n]; end for;
for i in [1..20], p in [11,13,19,23], r in [2,3,4,5,6] do
n := Random([1..100]); j := Random([1..20]);
assert IsDivisibleBy(Binomial(p^r*n,p*j)-Binomial(p^(r-1)*n,j),
  p^(r+3)); end for;

```

2.3. The prime power case $n = p^r$. The difference $(x + y)^{p^r} - (x^p + y^p)^{p^{r-1}}$ is already divisible by p^r in $\mathbb{Z}[x, y]$: For i coprime to p , the binomial coefficient $\binom{p^r}{i}$ is divisible by p^r . Otherwise, $i = pj$, and x^{pj} appears in the two terms with coefficient $\binom{p^r}{pj}$ and $\binom{p^{r-1}}{j}$, that cancel out mod p^r by Lemma 2.2. In turn, if j is coprime to p , each term is divisible by p^{r-1} in its own right. The result follows by induction on j .

We require an equation $(1 + tx)^{p^r} - 1 \equiv 0$ modulo the hypersurface F and the relations for $\mathcal{B}_{p^r}$ in order to define TO_{p^r} as a locally principal divisor in the hybrid group $G_{\mathcal{B},t}$. For this, write $(1 + tx)^n$ as a sum of pieces divisible by powers of tx . These are also divisible by complementary powers of p : for $n = p^3$, have

$$(2.8) \quad (1 + tx)^{p^3} = f_0 + p(tx)^{p^2} f_1 + p^2(tx)^p f_2 + p^3 tx f_3$$

where

$$\begin{aligned}
 f_0 &= 1 + (tx)^{p^3}, \\
 f_1 &= \frac{(1+(tx)^{p^2})^p - (1+(tx)^{p^3})}{p(tx)^{p^2}}, \\
 f_2 &= \frac{(1+(tx)^p)^{p^2} - ((1+(tx)^{p^2})^p)}{p^2(tx)^p}, \\
 f_3 &= \frac{(1+tx)^{p^3} - (1+(tx)^p)^{p^2}}{p^3 tx}, \\
 f_0, f_1, f_2, f_3 &\in \mathbb{Z}[t, x].
 \end{aligned}
 \tag{2.9}$$

Here the divisibility of the coefficients by powers of p comes from Lemma 2.2, and the divisibility by the complementary powers of tx come simply from the fact that each f_i is a polynomial in $(tx)^{p^{r-i}}$ with no constant term. In any case, the denominators are only there to get cancelled in the following tautology, that is obvious if you simply add together the numerators of f_0, f_1, f_2, f_3 .

The base ring B has quantities S_1, S_2, S_3 subject to relations P_1, P_2, P_3

$$P_1 = S_1 t^{p^3-p^2} + p, \quad P_2 = S_2 t^{p^3-p} + p^2, \quad P_3 = S_3 t^{p^3-1} + p^3.
 \tag{2.10}$$

They fit together in the hypersurface equation F for $\mathbb{T}\mathbb{O}_{p^3} \subset \mathbb{G}_{B,t}$:

$$F = x^{p^3} - S_1 x^{p^2} f_1 - S_2 x^p f_2 - S_3 x f_3.
 \tag{2.11}$$

The conclusion is the identity we need

$$(1 + tx)^{p^3} - 1 = t^{p^3} F + \sum_{i=1}^3 P_i (tx)^{p^{3-i}} f_i.
 \tag{2.12}$$

in the coordinate ring of $\mathbb{T}\mathbb{O}_{p^3}$. Here the $P_i = 0$ in B or everywhere locally in $\mathcal{O}_{\mathcal{B}_{p^3}}$, and the single equation F defines a hypersurface in the hybrid group scheme that is a sub-group scheme of order p^3 .

Our theme of computer sanity checks is relevant for the above equations (2.8)–(2.12) the calculations are tautologies, but a misplaced bracket turns them into worthless nonsense.

```
R<S1,S2,S3,t,x> := PolynomialRing(Integers(),5);
```

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f0 := 1+(t*x)^(p^3);
f1 := ( (1+(t*x)^(p^2))^p - (1+(t*x)^(p^3)) )
      div (p*(t*x)^(p^2));
f2 := ( (1+(t*x)^p)^(p^2) - ((1+(t*x)^(p^2))^p) )
      div (p^2*(t*x)^p);
f3 := ( (1+t*x)^(p^3) - (1+(t*x)^p)^(p^2) ) div (p^3*t*x);
(1+t*x)^(p^3)
eq f0 + p*(t*x)^(p^2)*f1 + p^2*(t*x)^p*f2 + p^3*t*x*f3;
P1 := S1*t^(p^3-p^2) + p;
P2 := S2*t^(p^3-p) + p^2;
P3 := S3*t^(p^3-1) + p^3;
Fp3 := x^(p^3) - S1*x^(p^2)*f1 - S2*x^p*f2 - S3*x*f3;
(1+t*x)^(p^3) - 1 eq
t^(p^3)*Fp3 + P1*(t*x)^(p^2)*f1 + P2*(t*x)^p*f2 + P3*(t*x)*f3;

```

The same for p^2 :

```

R<S1,S2,t,x> := PolynomialRing(Integers(),4);
for p in [2,3,5,7,11,13] do
f0 := 1 + (t*x)^(p^2);
f1 := ( (1 + (t*x)^p)^p - (1 + (t*x)^(p^2)) ) div (p*(t*x)^p);
f2 := ( (1+t*x)^(p^2) - (1 + (t*x)^p)^p ) div (p^2*t*x);
(1+t*x)^(p^2);
$1 eq f0 + p*(t*x)^p*f1 + p^2*t*x*f2;
P1 := S1*t^(p^2-p)+p;
P2 := S2*t^(p^2-1)+p^2;
Fp2 := x^(p^2) - S1*x^p*f1 - S2*x*f2;
((1+t*x)^(p^2) - 1) eq t^(p^2)*Fp2 + P1*(t*x)^p*f1 + P2*t*x*f2;
end for;

```

2.4. Alchemy polynomials AL_{p^r} for prime powers $n = p^r$. The raw aCLM polynomial for $n = p^r$ is

$$(2.13) \quad F_{p^r} = \frac{(x+y)^{p^r} - (x^p + y^p)^{p^{r-1}}}{p^r} \in \mathbb{Z}[x, y].$$

Here *raw* means that we take out the numerical factor p^r , but not yet the polynomial factors $x, y, x+y$ and others. If $r = 1$ this is the classical CLM polynomial, with the predictable factors of (1.1), according as $p \equiv 5$ or $7 \pmod{6}$. For $p = 2$ and $p = 3$, the only factors are respectively xy and $xy(x+y)$, leaving 1 as main factor. It is also clear that F_{p^r} is divisible by each F_{p^i} for $i < r$.

The next result dismantles the raw aCLM polynomial F_{p^r} for $r > 1$ as a product of the “predictable factors” xy , powers of $x+y$ and $x^2 + xy + y^2$, followed by factors coming from lower powers p^i with $1 \leq i < r$, times a *main factor*, our alchemy polynomial AL_{p^r} .

Although there are small differences in the run-up, and 2 is ever the even man out, we state the formula for AL_{p^r} all in one go.

Definition 2.4. For $p = 2$, set $U = (x+y)^2$ and $V = x^2 + y^2$. For every odd prime p , set

$$(2.14) \quad U = (x+y)^{p-1} \text{ and } V = (x^p + y^p)/(x+y) = \sum_{i=0}^{p-1} (-1)^i x^{p-i-1} y^i.$$

Now for $r \geq 2$, define the alchemy polynomial to be

$$(2.15) \quad AL_{p^r} = \frac{1}{p} \cdot \sum_{i=1}^p (U^{p^{r-1}})^{p-i} \times (V^{p^{r-1}})^{i-1}$$

The role of AL_{p^r} in the construction of $\mathbb{T}\mathbb{O}_n$ appears in Remark 2.6, together with a discussion of its conjectured irreducibility. Quite remarkably, V is the homogeneous cyclotomic polynomial for $2p$ (this holds also for $p = 2$); moreover the sum in (2.15) is the homogeneous cyclotomic polynomial for p evaluated at $U^{p^{r-1}}$ and $V^{p^{r-1}}$. The formula in (2.15) is straightforward to write out in computer algebra (without any division or factorisation), although it grows too rapidly for hand calculation. Setting yourself the task of writing out the binomial coefficients $\binom{p^r}{i}$ for all i , say for $p^r = 5^2$ or $13^2 = 169$ or $19^4 = 130231$ will make this point clear.

Theorem 2.5. *The raw aCLM polynomials F_{p^r} factors as a product of predictable factors times a cumulative product of the AL_{p^i} of Definition 2.4, as follows:*

$$(2.16) \quad \begin{aligned} p = 2 : & \quad F_{2^r} = xy \times \prod_{i=2}^r AL_{2^i} \\ p = 3 : & \quad F_{3^r} = xy(x+y)^{3^{r-1}} \times \prod_{i=2}^r AL_{3^i} \\ p \equiv 5 \pmod{6} : & \quad F_{p^r} = xy(x+y)^{p^{r-1}}(x^2+xy+y^2) \times \prod_{i=2}^r AL_{p^i} \\ p \equiv 7 \pmod{6} : & \quad F_{p^r} = xy(x+y)^{p^{r-1}}(x^2+xy+y^2)^2 \times \prod_{i=2}^r AL_{p^i} \end{aligned}$$

Remark 2.6 (Irreducibility). The word *alchemy* started out as a playful acronym for aCLM polynomial. However, in the construction of n , the alchemy polynomials AL_n live up to their name, with the power to transmute additive group schemes into multiplicative.

They are also “almost cyclotomic”: we saw in Theorem 2.5 that each AL_{p^r} for $r \geq 2$ comes by substituting the cyclotomic-like expressions U and V into the homogenised cyclotomic polynomial for p .

The irreducibility of the classical CLM polynomials is a long-standing conjecture. The cyclotomic polynomial Φ_n has roots the primitive n th root of 1, the generators of μ_n ; one shows that Frobenius lifting permutes these transitively, which proves that Φ_n is irreducible. The roots of AL_{p^r} do not form a group, so we don’t have a ready-made formula for how Frobenius lifting permutes them. Instead, the way the Tate–Oort group $\mathbb{T}\mathbb{O}_{p^r} \rightarrow \mathcal{B}_{p^r}$ works is that the roots of AL_{p^r} define the degenerations of the prime Jordan–Hoelder subgroup of $\mathcal{C}_p \subset \mathcal{C}_{p^r}$ to the inseparable group scheme α_p . (A simple model is the additive group $\mathbb{F}_p^+$ defined by $(x^p - x) \subset \mathbb{G}_a$ viewed in the family $x^p - \lambda \cdot x$ with λ tending to 0. The p elements of $\mathbb{F}_p^+ \cong \mathbb{Z}/p$ degenerate to a single point of multiplicity p forming the subgroup scheme $\alpha_p \subset \mathbb{G}_a$.) We believe that a careful discussion of the groups $\mathbb{T}\mathbb{O}_{p^r}$, and their characterisation (uniquely up to isomorphism with stated properties) would prove that these degenerations form a single conjugacy class under Galois extensions, with the conjectured irreducibility as a corollary.

2.5. The aCLM polynomial for general n . For $n = \prod_{i=1}^k p_i^{r_i}$ the product of k distinct primes p_i to powers $r_i \geq 1$, the square-free divisors are the 2^k products $a_I = \prod_{i \in I} p_i$ for $I \subset [1, \dots, k]$. Set $b_I = n/a_I$. Each polynomial $(x^{a_I} + y^{a_I})^{b_I}$ is homogeneous of degree n . The following lemma generalises the case $n = pq$ of Proposition 2.1:

Lemma 2.7. *The alternating sum of the $(x^{a_I} + y^{a_I})^{b_I}$ is divisible by n in $\mathbb{Z}[x, y]$:*

$$(2.17) \quad F_n = \frac{1}{n} \left(\sum_I (-1)^{\#I} (x^{a_I} + y^{a_I})^{b_I} \right) \in \mathbb{Z}[x, y].$$

This defines the *aCLM polynomial* $F_n(x, y)$; for $\mathbb{T}\mathbb{O}_n$, we work with the inhomogeneous form

$$(2.18) \quad f_n(x, t) = F_n(xt, 1)/t = t^{n-2}x^{n-1} + \dots + x.$$

For now, we do not cancel trivial factors from F_n or f_n . The literature is largely concerned with the conjectured irreducibility after cancellation, but we focus on the group scheme $\mathbb{T}\mathbb{O}_n$. The factor x provides the unit element $x = 0$ of the group or group scheme, and we prefer not to cancel it.

For examples ad lib, run the Magma code below.

```

R<x> := PolynomialRing(Rationals());
aLCM := func< n |
  &+[(-1)^(#Factorisation(a))*(1+x^a)^(n div a) : a in
    [a : a in Divisors(n) | IsSquarefree(a) ]] div n >;
[aLCM(n) : n in [8..17]];

```

with output

$$\begin{aligned}
(2.19) \quad F_8 &= x^7 + 3x^6 + 7x^5 + 8x^4 + 7x^3 + 3x^2 + x \\
F_9 &= x^8 + 4x^7 + 9x^6 + 14x^5 + 14x^4 + 9x^3 + 4x^2 + x \\
F_{10} &= x^9 + 4x^8 + 12x^7 + 20x^6 + 25x^5 + 20x^4 + 12x^3 + 4x^2 + x \\
F_{11} &= x^{10} + 5x^9 + 15x^8 + 30x^7 + 42x^6 + 42x^5 + 30x^4 + 15x^3 \\
&\quad + 5x^2 + x \\
F_{12} &= x^{11} + 5x^{10} + 18x^9 + 40x^8 + 66x^7 + 75x^6 + 66x^5 + 40x^4 \\
&\quad + 18x^3 + 5x^2 + x \\
F_{13} &= x^{12} + 6x^{11} + 22x^{10} + 55x^9 + 99x^8 + 132x^7 + 132x^6 \\
&\quad + 99x^5 + 55x^4 + 22x^3 + 6x^2 + x \\
F_{14} &= x^{13} + 6x^{12} + 26x^{11} + 70x^{10} + 143x^9 + 212x^8 + 245x^7 \\
&\quad + 212x^6 + 143x^5 + 70x^4 + 26x^3 + 6x^2 + x \\
F_{15} &= x^{14} + 7x^{13} + 30x^{12} + 91x^{11} + 200x^{10} + 333x^9 + 429x^8 \\
&\quad + 429x^7 + 333x^6 + 200x^5 + 91x^4 + 30x^3 + 7x^2 + x \\
F_{16} &= x^{15} + 7x^{14} + 35x^{13} + 112x^{12} + 273x^{11} + 497x^{10} + 715x^9 \\
&\quad + 800x^8 + 715x^7 + 497x^6 + 273x^5 + 112x^4 + 35x^3 \\
&\quad + 7x^2 + x \\
F_{17} &= x^{16} + 8x^{15} + 40x^{14} + 140x^{13} + 364x^{12} + 728x^{11} + 1144x^{10} \\
&\quad + 1430x^9 + 1430x^8 + 1144x^7 + 728x^6 + 364x^5 + 140x^4 \\
&\quad + 40x^3 + 8x^2 + x
\end{aligned}$$

proof of Lemma 2.7. In a word, we arrange the terms not divisible by n in the alternating sum of (2.17) to cancel out modulo powers of the p_i by repeated use of the Cancellation Lemma 2.2.

In more detail, work by induction, introducing one prime p_i at a time to the summands of (2.17). In the notation introduced before Lemma 2.7, for $j = 0, 1, \dots, k$ write

$$(2.20) \quad g_j = \sum_{I \subset [0, \dots, j]} (-1)^{\#I} (x^{a_I} + y^{a_I})^{b_I}.$$

When $j = 0$ we get the unadapted $g_0 = (x + y)^n$. This treats the empty set of prime factors of n , but the coefficient of the term $x^{n-i}y^i$ in g_0 is already divisible by n for any i coprime to n .

Next, $j = 1$ gives $g_1 = (x + y)^n - (x^{p_1} + y^{p_1})^{n/p_1}$. This does not modify any term $x^{n-i}y^i$ with $p_1 \nmid i$, but the coefficient of a term with $p_1 \mid i$ go from $\binom{n}{i}$ to $\binom{n}{i} - \binom{n/p_1}{i/p_1}$, which is divisible by $p_1^{r_1}$ by Lemma 2.2.

The case $j = 2$ was illustrated in 2.2. The inductive step from j to $j + 1$ adds the next prime p_{j+1} , and doubles the number of summands, that are indexed by the power set of $[0, \dots, j]$.

By induction, assume that every term $x^{n-i}y^i$ in g_j with i divisible by one of $p_1, \dots, p_j$ has coefficient divisible by the appropriate power of that prime. Passing from g_j to g_{j+1} only modifies terms $x^{n-i}y^i$ with i divisible by p_{j+1} . By construction, every coefficient of g_j is an alternating sum of binomial coefficients. Since we pass from the power set of $[1, \dots, j]$ to that of $[1, \dots, j+1]$ adding $j+1$, each summand of g_j is matched with a new summand with opposite sign. In this new summand (corresponding to some fixed $I \subset [0, \dots, j]$ and $I' = I \sqcup \{j+1\}$), each term $x^{n-i}y^i$ under modification has coefficient going from $\binom{b_I}{i/a_I}$ to $\binom{b_{I'}=b_I/p_{j+1}}{i/(a_I p_{j+1})}$, so the difference in the adapted g_{j+1} is divisible by $p_{j+1}^{r_{j+1}}$ by Lemma 2.2. This proves the inductive step. $\square$

3. CONSTRUCTION OF TO_n

3.1. Case $n = p$ a single prime. This is the model case from $[\mathrm{TOp}]$. Over the ground ring $\mathbb{Z}$, the *base ring* for TO_p is $B = \mathbb{Z}[S, t]/(P)$ where $P = St^{p-1} + p$. It is the subring of $\mathbb{Z}[t, t^{-1}]$ that allows t^{p-1} in the denominator for each power of p in the numerator. Here t is the hybrid parameter in $G_{B,t}$, that changes gear from additive to multiplicative. Recall the group law on $G_{B,t}$ is

$$(3.1) \quad m: \mathbb{A}^1 \times \mathbb{A}^1 \rightarrow \mathbb{A}^1 \quad \text{given by } (x_1, x_2) \mapsto x_1 + x_2 + tx_1x_2$$

Adjoining t rather than $T = t^{p-1}$ is a pullback by a cyclic Galois cover of order $p-1$. The coordinate ring of TO_p is $A = B[x]/(F)$ modulo the monic degree p polynomial $F = x^p - Sf_p(t, x)$ where

$$(3.2) \quad f_p(t, x) = \frac{(1+tx)^p - 1 - (tx)^p}{pt} = \sum_{i=1}^{p-1} \frac{\binom{p}{i}}{p} t^{i-1} x^i,$$

from the original CLM polynomial. This gives

$$(3.3) \quad \begin{aligned} p = 2 &: x^2 - Sx, \\ p = 3 &: x^3 - Stx^2 - Sx, \\ p = 5 &: x^5 - St^3x^4 - 2St^2x^3 - 2Stx^2 - Sx, \\ p = 7 &: x^7 - St^5x^6 - 3St^4x^5 - 5St^3x^4 - \dots \quad \text{and so on.} \end{aligned}$$

The point of these definitions is that $(1+tx)^p \equiv 1$ modulo (P, F) so that $A = \mathbb{Z}[S, t]/(P, F)$ has $\tau = 1+tx$ as a primitive p th root of 1. This allows us to define TO_p as the hypersurface $V(F) \subset G_{B,t}$, which is a p -torsion subgroupscheme because $\begin{pmatrix} 1 & 0 \\ x & 1+tx \end{pmatrix}^p = \mathrm{Id}$.

3.2. Case $n = pq$ the product of two primes. For distinct primes p, q , introduce S_p and $S_q \in \mathbb{Z}[t^{\pm 1}]$ subject to relations

$$(3.4) \quad \begin{aligned} P_p &= S_p t^{(p-1)q} + p = 0 \quad \text{and} \quad P_q = S_q t^{p(q-1)} + q = 0, \\ \text{that is} \quad S_p &= -p/t^{(p-1)q}, \quad S_q = -q/t^{p(q-1)}. \end{aligned}$$

The base ring is $B = \mathbb{Z}[t, S_p, S_q]/(P_p, P_q)$ with defining relations (3.4). We also introduce the notation

$$(3.5) \quad S_{pq} = -t^{(p-1)(q-1)} S_p S_q \in B \quad \text{with relation} \quad P_{pq} = S_{pq} t^{pq-1} + pq = 0.$$

As before, $B \subset \mathbb{Z}[t^{\pm 1}]$ adjoins S_p, S_q and S_{pq} , allowing specified powers of t in denominators provided that powers of p and q are in the numerator. Here S_{pq} is not needed as a generator of B , but $(p-1)(q-1) = \varphi(pq)$ is the order of the multiplicative group $(\mathbb{Z}/n)^\times = \text{Aut}(\mathbb{Z}/(pq))$. For example, $(\mathbb{Z}/15)^\times$ is $\mathbb{Z}/4 \times \mathbb{Z}/2$ generated by 2 and 7 with $2^4 = 1$ and $7^2 = 1$.

The coordinate ring of $\mathbb{T}\mathbb{O}_{pq}$ is $A = B[x]/(F)$ with defining polynomial

$$(3.6) \quad \begin{aligned} F &= x^{pq} - S_p f_p - S_q f_q - S_{pq} f_{pq}, \quad \text{where} \\ f_p &= \frac{(1 + (tx)^q)^p - 1 - (tx)^{pq}}{pt^q}, \quad f_q = \frac{(1 + (tx)^p)^q - 1 - (tx)^{pq}}{qt^p}, \\ f_{pq} &= \frac{(1 + tx)^{pq} - (1 + (tx)^p)^q - (1 + (tx)^q)^p + (1 + (tx)^{pq})}{pqt} \end{aligned}$$

The f_p, f_q and f_{pq} are our aCLM polynomials (2.3).

Theorem 3.1. *The definitions (3.6) give $A = B[x]/(F)$ the n th root of unity $1+tx$. In fact*

$$(3.7) \quad (1 + tx)^{pq} - 1 = t^{pq} F + t^p f_q P_q + t^q f_p P_p + t f_{pq} P_{pq}.$$

The powers $pq, p, q, 1$ on t and the subscripts on the f are complementary, multiplying to n .

Remark 3.2. *As a result, $\mathbb{T}\mathbb{O}_n = \text{Spec} A$ with $n = pq$ is a group scheme over B , with the multiplication (3.1). It is flat over B , and is a Galois twist of $\mathbb{Z}/n$ wherever S_p, S_q, S_{pq} are invertible (in particular, in characteristic coprime to pq).*

Over the closed set $V(p) \subset \text{Spec} B$ (where q is of course invertible), it is a group scheme over a base ring with parameters t and S_p . Where t is invertible it is isomorphic to the reductive subgroup $\mu_{pq} \subset \mathbb{G}_m$ via $\tau = 1 + tx$, which is isomorphic to μ_p times μ_q (an etale twist of $\mathbb{Z}/q$ in characteristic p). Where S_p is invertible (but $p = 0$), it is still an etale twist of $\mathbb{Z}/n$. (Note we can't have $t = 0$ since $p \neq q$, so the degenerations $\mu_p \rightsquigarrow \alpha_p$ won't happen) The same vice-versa with p, q exchanged.

In other words, $\mathbb{T}\mathbb{O}_{pq}$ almost contains the same information as the fibre product of $\mathbb{T}\mathbb{O}_p$ and $\mathbb{T}\mathbb{O}_q$ over $\mathbb{Z}[t]$. (This is morally correct, although it does not make complete sense as stated, since $\mathbb{T}\mathbb{O}_p$ and $\mathbb{T}\mathbb{O}_q$ are defined over different base rings.)

Proof. A simple verification: $t^{pq} F$ on the r.h.s. of (3.7) is a sum of 4 terms from (3.6), the first of which is $t^{pq} x^{pq}$. On substituting the definitions of S_p, S_q, S_{pq} in the remaining 3 terms, the factors p, q and powers of t in the numerators and denominators cancel out, leaving the numerators of f_p, f_q and f_{pq} from (3.6):

$$(3.8) \quad \begin{aligned} &(1 + (tx)^p)^q - (1 + (tx)^{pq}), \quad (1 + (tx)^q)^p - (1 + (tx)^{pq}) \\ &(1 + tx)^{pq} - (1 + (tx)^p)^q - (1 + (tx)^q)^p + (1 + (tx)^{pq}) \end{aligned}$$

When summing these, the first two brackets on the top line cancel the middle brackets on the bottom line. The three occurrences of $1 + (tx)^{pq}$ add to $-(1 + (tx)^{pq})$. Thus the 4 terms on the right of (3.7) add to the left-hand side. $\square$

Calculations. The proof was rather trivial, but any number of errors with signs or exponents are possible (in practice, more or less inevitable). We work by making every possible error before getting to the correct statement, so a sanity check in computer algebra is indispensable:

```

A<x,t,Sp,Sq> := PolynomialRing(Integers(),4);
for p in [1..30], q in [p+1..50] do
  if IsPrime(p) and IsPrime(q) then [p,q];
  Pp := p + Sp*t^((p-1)*q); Pq := q + Sq*t^(p*(q-1));
  Spq := -Sp*Sq*t^((p-1)*(q-1)); Ppq := p*q + t^(p*q-1)*Spq;
  fp := ((1+(t*x)^q)^p - 1-(t*x)^(p*q)) div (t^q*p);
  fq := ((1+(t*x)^p)^q - 1-(t*x)^(p*q)) div (t^p*q);
  fpq := ((1+t*x)^(p*q) + 1^(p*q)+(t*x)^(p*q)
          - (1+(t*x)^p)^q - (1+(t*x)^q)^p) div (t*p*q);
  F := x^(p*q) - Sp*fp - Sq*fq - Spq*fpq;
  (1+t*x)^(p*q) - 1 eq
    t*fpq*Ppq + t^p*fq*Pq + t^q*fp*Pp + t^(p*q)*F;
end if; end for;

```

3.3. Case when n is square-free. For $n = \prod_{i=1}^k p_i$ the product of k distinct primes p_i , the result in section 3.2 can be easily generalized.

Adjust the definition in 2.20 by letting $a_I = \prod_{i \in I} p_i$, $b_I = n/a_I$ and

$$(3.9) \quad g_J = (-1)^{\#J} \sum_{I \subset J} (-1)^{\#I} (x^{b_I} + y^{b_I})^{a_I}.$$

for $J \subset [1, \dots, k]$. Note $g_\emptyset = x^n + y^n$ and $g_{[1, \dots, k]} = (-1)^k n F_n$ where F_n is the n -th aCLM polynomial. The following lemma generalizes the observations in (3.8), saying that we have a nice formula for the family of g_J :

Lemma 3.3.

$$(3.10) \quad (x + y)^n = \sum_{J \subset [1, \dots, k]} g_J$$

Proof. For a fixed $I \subsetneq [0, \dots, k]$, we show that the summand $(x^{b_I} + y^{b_I})^{a_I}$ gets canceled out. It is clear that $(-1)^{\#I + \#J} (x^{b_I} + y^{b_I})^{a_I}$ is a summand of g_J if and only if $I \subset J$. For $i = 0, \dots, k - \#I$, there are $\binom{k - \#I}{i}$ many different sets of size $\#I + i$ containing I and their contribution to the summand $(x^{b_I} + y^{b_I})^{a_I}$ is precisely $\binom{k - \#I}{i} (-1)^{i + \#I}$. Just notice that

$$(3.11) \quad \sum_{i=0}^{k - \#I} (-1)^{\#I + i} \binom{k - \#I}{i} = (-1)^{\#I} (1 - 1)^{k - \#I} = 0$$

On the other hand, $(x + y)^n$ appears exactly once in $g_{[1, \dots, k]}$ on r.h.s. $\square$

Note that when n is square-free, a_I and b_I are in symmetric position. Therefore g_I is divisible by a_I by the inductive step in the proof of Lemma 2.7. Like aCLM polynomials, $g_I(1, x)$ has no constant term with lowest degree b_I . Define

$$(3.12) \quad f_I(t, x) = \frac{g_I(1, tx)}{a_I t^{b_I}} \in \mathbb{Z}[t, x]$$

Now introduce variables $S_I \in \mathbb{Z}[t^{pm1}]$ for $I \subset [1, \dots, k]$ subject to the relations

$$(3.13) \quad P_I = S_I t^{n-b_I} + a_I = 0, \text{ or } S_I = -a_I/t^{n-b_I}$$

The base ring is $B = \mathbb{Z}[t, \{S_I\}]/(\{P_I\})$ for $I \subset [1, \dots, k]$ with defining relations (3.13). The coordinate ring of $\mathbb{T}\mathbb{O}_n$ is $B[x]/(F)$ with defining polynomial

$$(3.14) \quad F = x^n - \sum_{I \text{ nonempty}} S_I f_I$$

Theorem 3.4. *These definitions give $\mathbb{T}\mathbb{O}_n$ the n -th root of unity $1 + tx$. In fact,*

$$(3.15) \quad (1 + tx)^n - 1 = t^n F + \sum_{I \text{ nonempty}} t^{b_I} f_I P_I$$

Proof. This can be easily verified as in theorem 3.1. Plugging in the definitions of P_I and f_I , this is reduced to lemma 3.3. $\square$

Note that we are essentially copying the procedures of constructing $\mathbb{T}\mathbb{O}_{pq}$, hence the properties of the resulting $\mathbb{T}\mathbb{O}_n$ should be close to those of $\mathbb{T}\mathbb{O}_{pq}$ illustrated in remark 3.2. It is flat over B and is a Galois twist of the cyclic group $\mathbb{Z}/n$ wherever the variables S_I are invertible. Over $V(p) \subset \text{Spec}(B)$ with $p \mid n$ and where t is invertible, it is isomorphic to the reductive subgroup μ_n via $\tau = 1 + tx$, which is isomorphic to μ_p times an etale twist of $\mathbb{Z}/\frac{n}{p}\mathbb{Z}$ since n is square-free. And we can't have $t = 0$ over the base scheme when $k \geq 2$. It should be viewed as the fibre product of $\mathbb{T}\mathbb{O}_p$ with $p \mid n$. Of course, this doesn't make sense because the base rings are different.

3.4. Case $n = p^r$, particularly p^3 .

General aim: Construct the base ring $B = \mathbb{Z}[\{S_d\}, t]$ and coordinate ring $A = B[x]/(F)$ such that $(1 + tx)^n - 1$ is zero in A . In other words, it is a linear combination of the defining relations of B and the single monic polynomial $F = x^n + \dots$. The terms $(1 + tx)^n - 1$ in x^i of degree i coprime to n have binomial coefficient divisible by n . We deal with these by adjoining an S -variable $S_n = -n/t^{n-1}$ to B . Swapping $n \mapsto S_n t^{n-1}$ in the coefficient of x^i makes it divisible by t^n , and we put the quotient into F .

To deal with the terms of $(1 + tx)^n - 1$ in x^i with $\gcd(n, i) > 1$, take the divisors d of n in some order compatible with divisibility, and introduce S -variables $S_d = -d/t^d$ to do the same swap-and-divide by t^n trick.

For a prime power $n = p^r$ the partially ordered set of divisors of n is single-track. Everything here is trivial.

Lemma 3.5. *For p^r a prime power, we have $(1 + x)^{p^r} = f_0 + \sum_{i=1}^r p^i x^{p^{r-i}} f_i$, where $f_0 = 1 + x^{p^r}$ and*

$$(3.16) \quad f_i = (1 + x^{p^{r-i}})^{p^i} - (1 + x^{p^{r-i+1}})^{p^{i-1}} \quad \text{divided by } p^i x^{p^{r-i}}$$

Each quotient f_i has $p^i - 1$ terms and constant term 1.

A concertina cancellation of terms makes the statement a tautology. The key point of the lemma is that the $p^r + 1$ terms of $(1 + x)^{p^r}$ break up as a sum of combinations divisible by $p^i x^{p^{r-i}}$ for $i = 0, \dots, r$.

To see what is going on, work out the case $n = p^3$. Then the lemma states $(1+x)^{p^3} = f_0 + px^{p^2}f_1 + p^2x^pf_2 + p^3xf_3$, where

$$(3.17) \quad \begin{aligned} f_0 &= 1 + x^{p^3} & f_1 &= \frac{(1+x^{p^2})^p - (1+x^{p^3})}{px^{p^2}} \\ f_2 &= \frac{(1+x^p)^{p^2} - (1+x^{p^2})^p}{p^2x^p} & f_3 &= \frac{(1+x)^{p^3} - (1+x^p)^{p^2}}{p^3x}. \end{aligned}$$

Lemma 2.2 gives the divisibility by p^i in the formula for f_i . Each numerator is a polynomial in $x^{p^{r-i}}$ with no constant term, so divisible by $x^{p^{r-i}}$.

In the formula for $(1+x)^{p^r}$, the multiple $p^i x^{p^{r-i}}$ cancels the denominator of f_i . After this, the second term of each summand cancels the first term of its predecessor, making the statement a triviality.

For the base scheme $\mathcal{B}_{p^r}$ of $\mathbb{T}\mathbb{O}_{p^r}$ and the equation F of $\mathbb{T}\mathbb{O}_n \subset G_{\mathcal{B},t}$, first set $B = \mathbb{Z}[S_1, t]/(P)$ with defining relation $P = S_1 t^{p^r - p^{r-1}} + p$, so that $S_1 = -p/t^{p^r - p^{r-1}} = -p/t^{\varphi(p^r)}$.

Introduce $S_i = -p^i/t^{p^r - p^{r-i}} \in B$ satisfying the relations

$$(3.18) \quad P_i = S_i t^{p^r - p^{r-i}} + p^i \quad \text{for } i = 1, \dots, r.$$

Each S_i is (± 1 times) the i th power of $S = S_1$ times a positive power of t . We use them to replace $p^i \mapsto -S_i t^{p^r - p^i}$ in the tautology

$$(3.19) \quad (1+tx)^{p^r} = \text{sum of terms divisible by } p^i x^{p^{r-i}}.$$

of Lemma 3.5. The S_i are already in the ring $B[S, t]/(P)$, and the relations P_i of (3.18) are already in the ideal generated by $P = P_1$, so the S_i and P_i do nothing for B , and are just convenient notation for what follows.

Theorem 3.6. *Keep f_i as in (3.16) and the S_i as above. Define*

$$(3.20) \quad F := x^{p^r} - \sum_{i=1}^r S_i x^{p^{r-i}} f_i.$$

This gives the identity in $\mathbb{Z}[S, t]$:

$$(3.21) \quad (1+xt)^{p^r} - 1 = t^{p^r} F + \sum_{i=1}^r P_i (xt)^{p^{r-i}} f_i.$$

The base ring $B = \mathbb{Z}[S, t]/(P)$ has the relations P_i of (3.18), and F_{p^r} is the single relation in $A = B[x]/(F_{p^r})$. The conclusion is that (3.21) gives $\tau = 1+xt \in A$ as an n th root of 1.

To continue my theme on computer sanity checks: Fortunately, there are still things in the world we can be certain about:

```
R<x,t,S1> := PolynomialRing(Rationals(),3);
for p in [2,3,5,7,11,13] do p^2;
P1 := S1*t^(p^2-p) + p; S2 := -S1^2*t^((p-1)^2);
IsDivisibleBy(S2*t^(p^2-1)+p^2, P1); // all true
f0 := 1+(t*x)^p^2;
f1 := ((1+(t*x)^p)^p - (1+(t*x)^p^2)) div (p*x^p);
f2 := (1+(t*x)^p^2 - (1+(t*x)^p)^p) div (p^2*x);
f0 + p*x^p*f1 + p^2*x*f2 eq 1+(t*x)^p^2;
end for;
```

Still the case $n = p^3$ is representative. The base ring B has quantities S_1, S_2, S_3 subject to relations P_1, P_2, P_3

$$(3.22) \quad P_1 = S_1 t^{p^3 - p^2} + p, \quad P_2 = S_2 t^{p^3 - p} + p^2, \quad P_3 = S_3 t^{p^3 - 1} + p^3.$$

The polynomial defining $\mathbb{T}\mathbb{O}_{p^3}$ is $F = x^{p^3} - S_1 x^{p^2} f_1 - S_2 x^p f_2 - S_3 x f_3$. The base is stratified according to which of the four variables t, S_1, S_2, S_3 is invertible or zero. In (3.22) S_3 invertible means F is separable of degree p^3 – so the fibre group scheme is $\mathcal{C}_{p^3}$ there.

Next, $S_3 = 0$ and S_2 invertible means F_{p^3} is separable of degree p^2 in x^p , i.e. $F_{p^3}(x) = G(x^p)$ with G separable of degree p^2 . Then have $\mathbb{T}\mathbb{O}_{p^3} \rightarrow \text{Spec}(B[y]/G(y))$ via $y \mapsto x^p$ which is surjective by the lying-over theorem. Also S_2 invertible implies $\text{Spec}(B[y]/G(y)) \cong \mathbb{T}\mathbb{O}_{p^2}[1/S_2] \cong \mathcal{C}_{p^2}$. The kernel is precisely μ_p . So the fibre group is μ_p times $\mathcal{C}_{p^2}$.

The next case is clearly $S_2 = S_3 = 0$ and S_1 invertible; then F_{p^3} is separable of degree p in x^{p^2} , so the fibre group scheme is μ_{p^2} times $\mathcal{C}_p$.

At the start of the chain $S_1 = S_2 = S_3 = 0$, the fibre group scheme is the nonreduced point $x^{p^3} = 0$. Where t is invertible it is μ_{p^3} . The single point $S_1 = S_2 = S_3 = t = 0$ is the point where the t -line meets the S_1 -line, that is the transition from μ_{p^3} via the extension $1 \rightarrow \mu_{p^2} \rightarrow G \rightarrow \alpha_p \rightarrow 0$ to $\mu_{p^2} \times \mathcal{C}_p$. The remaining two wall-crossings in the chain are analogous

3.5. Blueprint for general n . To develop the general case, we work by analogy with the Euler phi function φ . It does

$$(3.23) \quad n = \prod_{i=1}^k p_i^{r_i} \longmapsto \varphi(n) = n \cdot \prod (1 - \frac{1}{p_i}) = \prod (p_i - 1) p_i^{r_i - 1}.$$

The product $\varphi(n) = n \cdot \prod (1 - \frac{1}{p_i})$ has 2^k terms, a Venn diagram on integers according to divisibility by p_i . Divisibility by p_i and p_j are independent conditions (the Chinese remainder theorem), so φ counts integers in $\{1, \dots, n\}$ coprime to n . A residue mod n that is divisible by just one p_i is taken out once by $1 - \frac{1}{p_i}$. If it is divisible by exactly two, it is taken out twice by $1 - \frac{1}{p_i}$ and $1 - \frac{1}{p_j}$, but restored by the product. If divisible by 3 of the p_i it is taken out 3 times, restored 3 times, then taken out by the final term in the sum.

The remaining factors $p_i^{r_i - 1}$ assert that for $r_i \geq 2$ we only do this filtering once for each p_i , reflecting the *divided power exponential* isomorphism

$$(3.24) \quad (\mathbb{Z}/p^a)^\times \cong (\mathbb{Z}/p)^\times \oplus (\mathbb{Z}/p^{a-1})^+$$

An integer m that is a primitive root mod p is coprime to p , so has order p^{a-1} in $\mathbb{Z}/p^{a-1}$.

Our aCLM polynomials make all the binomial coefficients in $(x + y)^n$ divisible n by a similar process.

Remark 3.7 (Further outlook). *When we have it all properly set up, and have time to work out Cartier duality in detail (this has some tricky details, as in [TOp, Section 4], but it can certainly be done), our $\mathbb{T}\mathbb{O}_{p^r}$ will allow any prime Jordan–Hoelder subquotient of $\mathbb{Z}/p^r$ to degenerate down to α_p and then out to μ_p , linking all*

$$(3.25) \quad \mu_{p^i} \times C_{p^{r-i}}$$

in a chain. (or separable forms of the $\mathbb{Z}/p^b$) for each $a + b = r$ with Cartier dual

$$(3.26) \quad \mu_{p^{r-i}} \times C_{p^i}$$

Viewed as degenerations of $\mathbb{Z}/p^n$, each subgroup $\mathbb{Z}/p^a \subset \mathbb{Z}/p^r$ may contract down to the nonreduced point group scheme α_{p^a} , corresponding to a stratification of the base scheme. This stratified deformation of AL_{p^n} is a kind of analog of the universal unfolding of the simple A_n singularity $\sum x_i^2 + y^{n+1}$ with successive strata representing $\sum x_i^2 + y^{n+1} + u_m y^{m+1}$ giving A_m for $m \leq n$.

In a group scheme, the connected component of the origin is of course a normal subgroup scheme, with the other components as cosets. For degenerations of $\mathbb{Z}/p^r$ (or its form), the connected component can only be AL_{p^i} for some $i \in [0, \dots, r]$, with the quotient a form of $\mathbb{Z}/p^{r-i}$.

The degenerations of μ_{p^r} are dual to those for $\mathbb{Z}/p^r$. In this case μ_{p^r} is connected, so arguing on the connected component gives nothing. The first map $\mu_{p^i} \hookrightarrow \mu_{p^r}$ of (3.26) can be seen as

$$(3.27) \quad \text{Spec}(B[y]/(y^{p^i} - 1)) \hookrightarrow \text{Spec}(B[x]/(x^{p^r} - 1))$$

given by setting $y = x^{p^{r-i}}$, and the second map is group multiplication by p^i in $\mu_{p^r} \subset \mathbb{G}_m$. This deformation is still a subgroup of the hybrid $G_{B,t}$ for appropriate base data B, t .

Ideally, $\mathbb{T}\mathbb{O}_n$ should consist of a web of cyclic group schemes of order n in a deformation family over the arithmetic toric base scheme $\mathcal{B}_n$. However, our current method does not reach this goal for general n because the constructions of $\mathbb{T}\mathbb{O}_{p^r}$ and $\mathbb{T}\mathbb{O}_n$ with n square-free are not compatible. Nor is $\mathbb{T}\mathbb{O}_{pq}$ as close to a genuine fibre product as we would like: over $p = 0$, the μ_p part of the fibre group scheme does not degenerate as it does for $\mathbb{T}\mathbb{O}_p$ because $V(t) = \emptyset$ in $\mathcal{B}_{pq}$, so the “wall-crossing” phenomenon does not occur there as we desired in remark 3.2.

This suggests that in general $\mathbb{T}\mathbb{O}_n$ cannot be realized inside a single ambient hybrid group: our original approach of defining it as a hypersurface cut out by one polynomial is too rigid. Instead, we need a separate hybrid group construction over each “wall-crossing” stratum and glue them appropriately, and this is beyond the scope of the present paper.

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